Fiscal Year:	FY 2005	**Task Last Updated:**	FY 03/29/2006
PI Name:	Nelson, Gregory A. Ph.D.		
Project Title:	NSCOR: Progressive Alterations of Central Nervous System Structure and Function Are Caused by Charged Particle Radiation		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	grnelson@llu.edu	**Fax:**	FY 909-558-4035
PI Organization Type:	UNIVERSITY	**Phone:**	909-558-8364
Organization Name:	Loma Linda University		
PI Address 1:	Basic Sciences, Div. Radiation Research 11175 Campus Street		
PI Address 2:	Chan Shun Pavilion, Room A-1024		
PI Web Page:			
City:	Loma Linda	**State:**	CA
Zip Code:	92350-1700	**Congressional District:**	31
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NSCOR 03-OBPR-02
Start Date:	10/01/2003	**End Date:**	09/30/2008
No. of Post Docs:	5	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Archambeau, John O. (Loma Linda University) Chang, Polly Y. (SRI International) Fike, John R. (University of California San Francisco) Krucker, Thomas (Novartis Institute for Biomedical Research) Lin, Weili (University of North Carolina at Chapel Hill) Mao, Xao-Wen (Loma Linda University) Obenaus, Andre (Loma Linda University) Pecaut, Michael J (Loma Linda University) Song, Sheng-Kwei (Washington University School of Medicine) Spigelman, Igor (University of California at Los Angeles)		
Grant/Contract No.:	NNJ04HC90G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The experimental program detailed above will provide a comprehensive, quantitative assessment of the effects of charged particle radiation on the mammalian brain using the mouse hippocampus as a model. The mouse brain will be irradiated with accelerated protons and iron ions and examined over the course of two years for progressive changes in hippocampal structure and function. Cell populations in tissue sections will be quantified by a high precision stereology method to measure rates of loss of neurons, vascular endothelial cells and glia. Brain blood vessel topology will be further characterized by an erosion casting technique that produces a 3-D map of the vessels at 1 micron resolution. Stem cells and ongoing neurogenesis will be quantified using a complementary technique in which dividing cells are directly detected. Activation of cells and residual genetic damage in these cells will be characterized by immunofluorescence and mRNA level measurements of selected cytokine biomarkers and analysis of mutations in the transgene b-galactosidase. Functional impairments of neuronal ensembles will be quantified by electrophysiological measurements of hippocampal tissue slices and patch clamp measurements will detail changes in membrane properties. System level changes will be quantified by electroencephalography and two magnetic resonance imaging modalities that map white matter tracts (diffusion tensor imaging) and local metabolic rate (oxygen extraction). The plasticity of the brain as a system will be tested in its reaction to an immune system challenge mediated by lipopolysacharide. Finally the ability of radiation to modulate the latency of a neurodegeneration process will be tested in a transgenic mouse that overproduced amyloid precursor protein and develops an Alzheimer disease-like pathology. Taken together these results should provide a basis for determining risk to the CNS from charged particle radiation exposure. The main hypothesis for the grant is: Charged particle radiation exposure induces progressive deleterious structural and functional changes in the brain, leading to performance decline. Research Aims: Hypothesis (A) Charged particle irradiation causes a progressive loss of cells and a remodeling of CNS tissue as a function of dose, time and LET. To test hypothesis A we will: Specific Aim A1. Quantify the cellular composition of mouse brain vascular endothelium using stereological analysis of preserved tissue. Specific Aim A2. Quantify the composition of mouse neurons, selected glia and stem cells using stereological analysis of preserved tissue. Specific Aim A3. Quantify the activation of resident and infiltrating mouse immune cells using stereological analysis immune-labeled tissue. Specific Aim A4. Quantify key molecular biomarkers associated with radiation-induced damage to mouse cells and their microenvironment using targeted gene expression profiling and mutation analysis of a structural gene. Hypothesis (B) Charged particle irradiation alters the functional output or performance of the brain as a function of dose, time and LET. To test hypothesis B we will: Specific Aim B1. Quantify macroscopic electrical properties of mouse brains using electro-encephalography (EEG). Specific Aim B2. Quantify tissue-level electrophysiological properties of mouse brain using tissue slices that represent intact regional ensembles of interacting cells. (extracellular, patch clamp) Specific Aim B3. Map the time evolution of macroscopic structural features and metabolic alteration in mouse brains using MRI. Hypothesis (C) Exposure to radiation decreases disease latency and alters response to acute stressors. To test hypothesis C we will: Specific Aim C1. Quantify the magnitude and time course of electrophysiological changes in brain slices of mice expressing a human transgene that produces an Alzheimer disease-like pathology and limits output. Specific Aim C2. Quantify the electrophysiological performance of brain slices in mice reacting to the application of systemic immune stress using lipopolysaccharide as a surrogate infectious agent Foremost in the Critical Path Roadmap system for radiation effects on the nervous system is Risk # 39 - Damage to the Central Nervous System from Radiation Exposure. The highest priority questions identified with this risk are: 10.01 Are the biological effects of protons above 10 MeV sufficiently similar to photons that photon data can be used for their consequences? 10.04 Are their differences in response to particles with similar LET, but with different atomic numbers and energies? 10.05 Are there unique biological effects associated with HZE's? 10.07 How can animal and cell experiments be done and data best be used to extrapolate to the human risk from space radiation? 10.10 What are the risks from SPE's and what is their impact on operations, EVAs and surface exploration? and 10.12 What are the effects of age, gender, and inter-individual diversity? CPR radiation risks 38, 40, 41 and 42 pose similar questions. A recent NASA workshop (10) identified supporting questions to assist in designing experiments to elucidate central nervous system (CNS) effects in response to radiation exposure. These questions include: 1) What is the latency of CNS injury? 2) What cell types are involved in CNS injury (axons, neurons, glia)? and 3) What vascular changes occur in response to exposure. We have used this framework to focus our investigation on the effects of charged particles on the CNS. Specifically we will quantify cellular and functional changes in the hippocampus following high energy proton (main constituent of SPE's; Q's 10.01, 10.10) and 56Fe particle exposure (HZE's of different LET; Q's 10.04, 10.05). We will use male rodents with sufficient statistical samples to address inter-individual variation (Q 10.12) and will address workshop questions 1 through 3 by examining the responses of the different constituent cells of brain tissue and the latency of damage. Our strategy is therefore aligned with the programmatic goals set forth by NASA and its advisory groups.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The response of the central nervous sytem to ionizing radiation will have direct impact on the development of radiotherapy using charged particles. The ability of protons, helium ions and carbon ions to be localized precisely in treatment fields close to critical structures (eg. optic nerve) make charged particle radiotherapy highly desireable. The limiting biological constraint on such treatments is the reaction of normal tissue surrounding a tumor site. The results of the NSCOR task will quantify cell loss and tissue functional changes form proton and iron irradiation that can be used directly to estimate normal tissue complication probabilities in the context of radiotherapy.
	Start-up Activities Although the contractual start date is listed above as 10-1-2003 the funds to initiate the NSCOR were first available in February of 2004. Using pre-award authorization, grant activities began on January 19, 2004 with an all-hands meeting held at Loma Linda University at which team members conducted a three-day tutorial and workshop to educate each other on the plans and technical approaches to be used in joint experimental activities over the next several years. A systematic review of all techniques, reagents and equipment resulted in standardized protocols approved by the NSCOR directors. The external Science Advisory Committee was also able to meet and make recommendations on technical approaches, staffing and management approaches. Consortium agreements between Loma Linda University and the co-investigators' institutions were then established by the various business offices between January and late March such that significant technical activities could begin in April 2004. All teams then conducted validation tests of all protocols prior to the commencement of animal irradiation. Institutional Animal Care and Use Committee (IACUC) reviews and approvals were obtained by all participating institutions. Beam time proposals and supporting documents were prepared for Brookhaven National Laboratory (BNL) to secure access to the NASA Space Radiation Laboratory (NSRL) and beam time was approved for use of 600 MeV/n iron ions. NASA officials requested that BNL develop the 600 MeV/n iron beam to supplement 1000 MeV/n iron beams that were then

currently available. NSRL scheduling enabled the first animals to be irradiated with iron ions during the NSRL-4 experimental campaign in September, 2004. Thus, at the time of this report the most advanced ages of iron-irradiated experimental subjects are 10 months of a planned series of ages ranging up to 24 months. Complementary proton irradiations have also been conducted. Technical Progress Highlights As the series of subjects exhibiting appropriate age vs. dose characteristics develops we have commenced histological, biochemical, imaging, and electrophysiological measurements. Unirradiated control subjects have also been used to establish baseline measurements and to validate protocols and intrinsic variability between subjects. At this time the small number of subjects analyzed and their apportionment by age and dose precludes any statistically significant conclusions to be made but preliminary observations indicate that we are able to measure the parameters that are important and that we have detected changes in the brains of irradiated animals as anticipated in the experimental design. Highlights and examples of data acquired are presented below.
Core activities: Irradiation techniques have been standardized with the use of stereotactic positioners for animals held under anesthesia in a system that pre-aligns mouse brains with collimator openings. Beam transport simulations and dosimetric measurements have been made with 1000 MeV/n and 600 MeV/n iron beams to characterize the irradiation conditions and a complementary collimator system for use with proton beams continues to be optimized and characterized. The system is comprised of an anesthesia/stereotactic holder for isoflurane anesthesia, quick disconnect gas connections and a three-part collimator . Personnel associated with all three hypotheses are now trained on the use of this system and are qualified users at the NSRL facility. In addition, four persons, Roman Vlkolinsky, Ph.D., Scripps Research Institute electrophysiologist, Shu-Wei Sun, Ph.D. Washington University MRI physicist, and LLU technical staff members Tamako Jones and Ella Lloyd have received advanced radiobiology training at BNL as part of the 2004 NASA Space Radiation Summer School Substantial progress has been made in establishing a database system at Loma Linda for archiving and cross referencing of all data sets generated by the NSCOR. Mr. Brad Karain has led this information technology effort which included setting up of a multi-terabyte server system, MySQL software-based database development using the Linux operating system. Data in formats ranging from MRI images to electrophysiological recordings to histological photographs and physical dosimetry are now entered into the database and indexed to individual animal subjects and time-dose cohorts.

Hypothesis A Activities. Immunohistochemical methods are now standardized for identification of specific cell types. Confocal microscope images of hippocampal sections labeled for atrocytes and microvessel-associated pericytes are being assembled into sequential stacks by stereology software developed for use on the Biorad MRC2400 confocal microscope system. Bromodeoxyuridine incorporated into the newly-synthesized DNA of dividing neuronal stem cells in subgranular layer of hippocampus can be visualized by anti-BrdUTP antibodies to quantified newly- born neurons cells and assess the ability of the irradiated brain to repopulate damaged regions. The cells later migrate and integrate into the hippocampal circuitry. Tissue from certain time and dose points for irradiated lacZ-gene containing transgenic mice has been archived for DNA extraction and mutation analysis. Plans for continued irradiation of mice with iron ions and protons are in place for the second year of the project.

Task Progress:

Hypothesis B Activities. Three electrophysiology rigs with common architecture and components have been set up for hippocampal slice recordings at LLU, Scripps Research Institute, and U.C.L.A. Input-output curves, long term potentiation and other physiological measurements on control and irradiated animals have been acquired for a limited number of subjects as the cohort ages. Preliminary analysis of recordings show dose-dependent changes in intrinsic nerve electrical properties and in long term potentiation (simple learning paradigm) reflecting capacity for synaptic remodeling following iron ion irradiation. Similar observations were obtained from Hypothesis C using transgenic and LPS-treated animals. Standard magnetization sequences (T1 and T2 relaxations) and diffusion tensor images have been obtained for mouse brains using the Loma Linda Brucker MRI systems (4.7 T and 11.7T). White matter tracts (which constrain fluid diffusion along their longitudinal dimension) can be mapped in three dimensions. Magnetization sequences were developed at Washington University in St. Louis for implemented on the Loma Linda Brucker MRI system so that animal shipping and handling would be minimized. Data acquisition has begun to examine possible demyelination or restructuring of fiber tracts in irradiated animals. Similarly, oxygen extraction fraction sequences have been improved by the U.N.C. group and transformed for use on the Loma Linda Brucker MRI systems.

Hypothesis C Activities. Corrosion casts of control and APP23 mouse brains have been obtained by T. Krucker et al. enabling us to map the three dimensional topology of blood vessels. Regions of brain with and without infarcts caused by Amyloid plaque accumulation can be mapped for infarct frequency vs. age and image processing improvements now permit hippocampal areas to be specifically highlighted. 120 APP23 animals have been irradiated as of 4/7/05 and 138 C57Bl for LPS immune stressor treatments. We have found that the APP23 animals are somewhat fragile and do not ship well, leading to some losses following BNL irradiations. Most of these animals are scheduled for electrophysiological measurements with the majority of corrosion cast animals scheduled for years 3 and 4. Induction of inflammatory response by lipopolysaccharide (LPS) treatment has been visualized by glial cell activation marker CD68 in immunofluorescent preparations using antigen IBA1 to identify glial cells. The strategy of identifying activation markers has been substituted for initial plans to measure soluble cytokines. Electrophysiological measurements following LPS treatments show that LPS-induced inhibition of long term potentiation occurs as expected. Irradiated animals additionally show decreased dendritic excitability and reduced synaptic efficacy in the short term. Data suggest that these changes may recover after 4 months. As can be seen from the examples above, the various methods are in place for carrying out the experimental plan and the NSCOR team is actively acquiring data from irradiated animals which must be maintained for up to 2 years post-irradiation before tissues are harvested for analysis. Irradiations will continue to populate the necessary time-dose cohorts for analysis of tissue structure and performance from 30 days to 2 years post exposure. Arrangements for the next round of iron irradiations are in place for October 2005 at NSRL.

Presentations

Nelson, G.; Archambeau, J.; Chang, P.; Fike, J.; Krucker, T.; Lin, W.; Mao, X.; Obenaus, A.; Pecaut, M.; Song, S.; Spigelman, I. "Progressive alterations of central nervous system structure and function are caused by charged particle radiation. " 16th Annual NASA Space Radiation Investigators' Workshop. Port Jefferson, N.Y., May 15-18, 2005.

Nelson, G.; Archambeau, J.; Chang, P.; Fike, J.; Krucker, T.; Lin, W.; Mao, X.; Obenaus, A.; Pecaut, M.; Song, S.; Spigelman, I. "Progressive alterations of central nervous system structure and function are caused by charged particle radiation. " Bioastronautics Investigators' Workshop. Galveston, TX. Jan. 10-12, 2005.

Nelson, G.; Archambeau, J.; Chang, P.; Fike, J.; Krucker, T.; Lin, W.; Mao, X.; Obenaus, A.; Pecaut, M.; Song, S.; Spigelman, I. "Progressive alterations of central nervous system structure and function are caused by charged particle radiation. " Presentation F2.2-0012-04. 35th COSPAR Assembly, Paris. Jul. 19, 2004.

	Nelson, G.; Archambeau, J.; Chang, P.; Fike, J.; Krucker, T.; Lin, W.; Mao, X.; Obenaus, A.; Pecaut, M.; Song, S.; Spigelman, I. "Progressive alterations of central nervous system structure and function are caused by charged particle radiation. " . 3rd International Workshop on Space Radiation Research. Port Jefferson, N.Y. May 16-20, 2004. Obenaus, A.; Robbins, M.; Kendall, E.; Smith, A.; Nelson, G. "Diffusion weighted imaging and magnetic resonance spectroscopy of the central nervous system after high LET radiation: temporal evolution." 16th Annual NASA Space Radiation Investigators' Workshop. Port Jefferson, N.Y., May 15-18, 2005 May-2005
Bibliography Type:	Description: (Last Updated: 03/26/2026)
Articles in Peer-reviewed Journals	Nelson, G.; Archambeau, J.; Chang, P.; Fike, J.; Krucker, T.; Lin, W.; Mao, X.; Obenaus, A.; Pecaut, M.; Song, S.; Spigelman, I. "Progressive alterations of central nervous system structure and function are caused by charged particle radiation." Radiation Research in press, September 2005. , Sep-2005